

TrES-3b

R-2338

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_120428 · created 2026-08-18T23:58:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456045.919 +/- 0.0009 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.021
Transit depth (Rp/Rs)^2	3.85 +/- 0.81 [%]
Orbital Inclination (inc)	81.35 +/- 0.41
Airmass coefficient 1 (a1)	0.9251 +/- 0.0083
Airmass coefficient 2 (a2)	0.074 +/- 0.028
Scatter in the residuals of the lightcurve fit is	0.9 %
Best Comparison Star	#3 - [324, 79]
Optimal Aperture	3.24
Optimal Annulus	5.52
Transit Duration (day)	0.054 +/- 0.0049

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.021 vs 0.16309 (deviation 1.45σ)
Duration fit vs published	0.054 d vs 0.0567 d (deviation 0.5σ)
Mid-time O–C	-2.7 min
Depth SNR	8.6
Residual scatter	0.9 %

Light curve

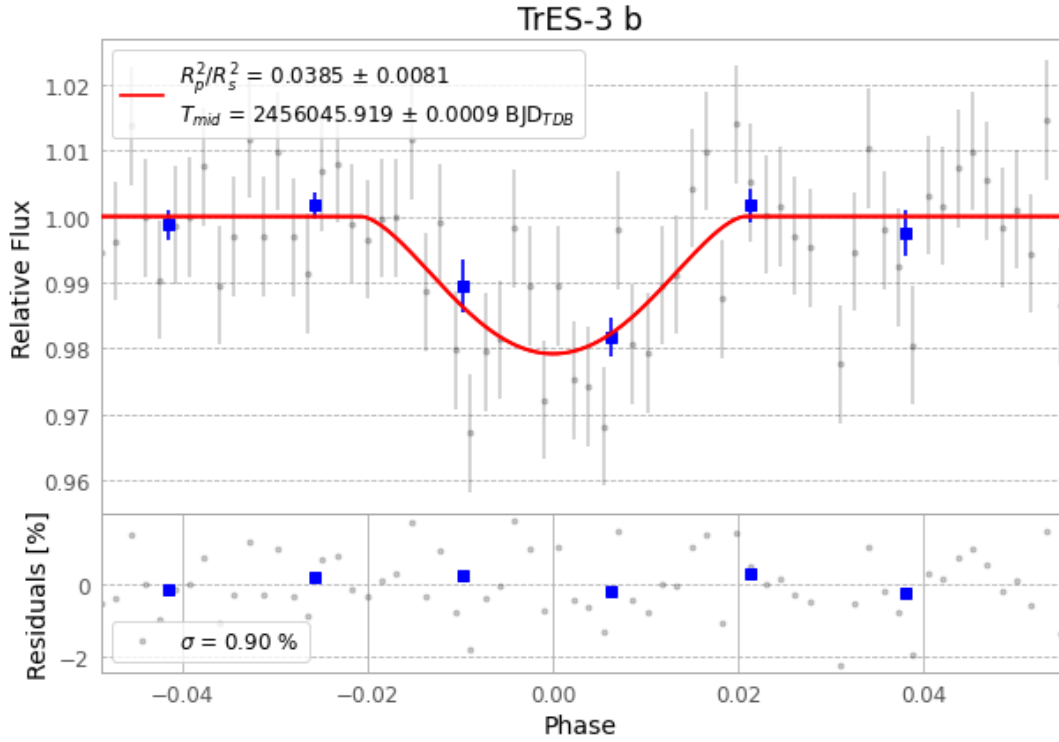


Figure 1 — FinalLightCurve_TrES-3 b_2012-04-28.png (SHA-256 567c6cfd108459ef...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [395, 237], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8256e7a71ec33e96...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

66 FITS frames (43 MB), TRES-3120428082712.FITS → TRES-3120428114212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-120428.log (bbd3f01717ab...), FinalLightCurve_TrES-3 b_2012-04-28.png (567c6cfd1084...), FinalLightCurve_TrES-3 b_2012-04-28.csv (4355ad663286...)

Compute resources

Wall time 128.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-18 23:58Z.